RV Educational Institutions
RV College of Engineering

Autonomous Institution Affiliated to Visvesvaraya Technological University, Belagavi

Approved by AICTE, New Delhi

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Academic Year 2021-22 (Even Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

(Common to BT, CS, EC, EEE, ET & ISE)

Date	15/06/2022	Maximum Marks	10 + 50
Course Code	18HEM61	Duration	20 + 90 Min
Sem	VI Semester	CIE - II	

INTRODUCTION TO MANAGEMENT AND ECONOMICS

Note: Answer all the Questions.

PART - A

Q. No.1	Questions	M	BT	CO
1.1	Mention the factors considered while framing vision and mission.	2	L1	CO2
1.2	List the steps involved in strategic management process.	2	L1	CO2
1.3	Distinguish between organizational structure and organizational design.	2	L1	CO2
1.4	Define and list the characteristics of chain of command.	2	L1	CO2
1.5	Write a brief note on price level and price index.	2	L1	CO4

PART - B

Q. No.	Questions	M	BT	CO
2.	Imagine a 7-star hotel being established in the heart of a city. Prepare vision and mission for the hotel, perform SWOT analysis and formulate TWO strategies for the same.	10	L6	CO2
3.	Examine the cash flow amongst those business units that produce cash versus those that absorb cash for a firm using Boston Consulting Group (BCG) Matrix.	10	L4	CO2
4.	Exhibit an organizational structure for a multi-national automobile company using all possible types of departmentalization.	10	L3	CO2
5.	In today's contemporary world, most organizations are transforming from a taller structure to a flatter structure. Elucidate the given statement using the concept of span of control.	10	L5	CO2
6.	Discuss the effects of GDP and its components on the nation's economy in the post-pandemic environment using an appropriate example.	10	L2	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	--	08	--	02	10	--	--	--	--	--
	Test	Max Marks	--	40	--	10	--	10	10	10	10	10

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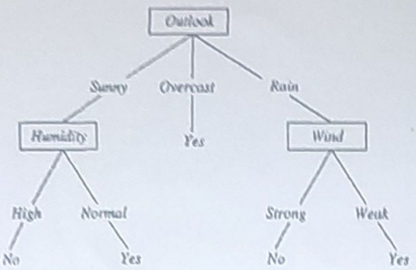
	R V College of Engineering **Department of Computer Science and Engineering** **CIE 2: Question Paper**		
Subject : (Code)	**Artificial Intelligence & Machine Learning (18CS62)**	**Semester:**	**VI**
Date : 15 June 2022	**Duration :** 120 minutes	**Staff: Dr Sindhu/ RESA/PH/Prof Merin**	
Name :	**USN :**	**Section :**	**A/B/C/ISE**

Q.No	Part-A	Marks	Level	CO
1	Learned trees from a decision tree can also be re-represented as sets of ______________ to improve human readability	01	1	3
2	Consider the decision tree given below  Represent this as a disjunction of conjunctions	01	3	1
3	What is the approximate inductive bias of ID3 ?	01	2	2
4	Give any 2 disadvantages of Instance based Learning	01	2	1
5	Why are lazy learning techniques called so ?	01	1	2
6	How are instances represented in case based reasoning ?	01	1	3
7	Give an expression for $\pi *$ or optimal policy	01	2	2
8	There are several approaches to avoiding over fitting in decision tree learning. Identify the 2 classes of such approaches?	01	3	2
9	Suggest any 2 conditions under which the Q learning algorithm converges towards a $\hat{Q}$ equal to the true Q function	02	2	4
Q. No	**PART B**			
1a	Use a prototypical example for a case based reasoning system and explain its working	5	2	1
1b	Explain locally weighted linear regression .	5	1	2
2 a	Give the steps in Q-learning algorithm and explain its relevance	4	2	2

2 b	Explain the learning task in the case of reinforcement learning with the help of MDP and relevant expressions	6	3	2
3 a	Use relevant examples and demonstrate which kind of problems are appropriate for decision tree learning	5	2	3
3 b	What are Radial Basis Functions? Draw a representative diagram for a Radial Basis Function network .	5	1	2
4	Use KNN Classifier to find the class of Score1 = 4.75 and score2 =3 . Assume k=3. Compare the results when k=5 (see table below). How are instance based techniques different from the other methods? How does the instance based method works when a new query instance is encountered?	10 (7 + 3)	3	4
5	Construct a decision tree for the following dataset . Use the concept of Entropy and Information gain to find the best attribute for split . (see table below)	10 (3 + 7)	4	4

Question 4 table:

Sl No	Score1	Score2	Class
1	8.25	3.25	C1
2	6	1.5	C2
3	4.25	3.75	C2
4	7.25	5.75	C1
5	4	4	C2
6	4.5	5	C2
7	3	3.25	C2
8	4.75	6.25	C1
9	3.5	5.25	C2
10	5.5	6.75	C1

Question 5 table:

Patient ID#	Sore Throat	Fever	Swollen Glands	Congestion	Headache	Diagnosis
1	Yes	Yes	Yes	Yes	Yes	Strep throat
2	No	No	No	Yes	Yes	Allergy
3	Yes	Yes	No	Yes	No	Cold
4	Yes	No	Yes	No	No	Strep throat
5	No	Yes	No	Yes	No	Cold
6	No	No	No	Yes	No	Allergy
7	No	No	Yes	No	No	Strep throat
8	Yes	No	No	Yes	Yes	Allergy
9	No	Yes	No	Yes	Yes	Cold
10	Yes	Yes	No	Yes	Yes	Cold

300 240
80
320

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DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING

Date	JUNE 2022	Maximum Marks	60
Course Code	18CS63	Duration	110 Min
Sem	VI Semester	CIE-II	
COMPILER DESIGN			

PART A

Sl.no		Marks	* L1-L6	*CO
1.1	State the difference between SLR and LR(1) parsing? What changes when constructing an LR(0) table versus an SLR table?	2	L2	CO3
1.2	Give an example for LALR parser generator.	1	L1	CO2
1.3	Construct a DAG for the expression x=a+(a+a+(a+a+a)+a)	2	L2	CO2
1.4	Given the following lex rules: is|am|are {printf("Verb\n");} island {printf("Island\n");} [a-zA-Z]+ {printf("Unknown\n");} How does lex choose *island* instead of is when it sees it?	2	L3	CO3
1.5	Explain the working principle of YACC with a block diagram.	2	L1	CO3

1.6	x = u - t;
	y = x * v;
	x = y + w;
	y = t - z;
	y = x * y;
	The minimum number of total variables required to convert the above code segment to static single assignment form is ______
		1	L2	CO1

PART B

Sl.no		Marks	* L1-L6	*CO
2.	Consider the given grammar. Construct the LALR parsing table for the grammar. S-> AA A->aA | b Parse the string "abb".	10	L3	CO4
3a.	Write a YACC Program to recognize a valid arithmetic expression and identify the identifiers and operators present. Print them separately.	4	L2	CO2
3b.	Show the steps does compiler take to resolve the S/R or R/R conflicts for the following ambiguous grammar E-> E + E | E * E | (E) | id	6	L2	CO4
4	Consider the following expression: X= (x+y)*(y+z)+(x+y+z) a) Write three-address code and its quadruple and triple representation b) Construct directed acyclic graph.	3+2+2+3	L3	CO4
5.	Consider the following (already augmented) grammar, which is known to be LR(1): S → Z Z → aMa | bMb | aRb | bRa M → c R → c Draw the LR(1) Automaton . Determine the grammar is in SLR or not.	10	L3	CO4
6a.	Describe the relationship between a production and an item in an LR(0) grammar. How does this relate to the notion of the stack in an LR(0) grammar? Give an algorithm for constructing the closure of a set of items LR(0) with respect to a particular grammar.	5	L3	CO2
6b.	Construct the set of LR(0) states and SLR parse table for this grammar and indicate whether the grammar is ambiguous. S -> A$ A -> -- A B | ++ A B | id B , B -> -- B | ++ B | ∈	5	L4	CO3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	1	12	11	36	3	15	37	5		

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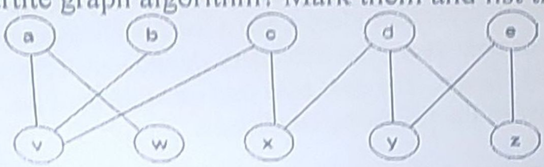
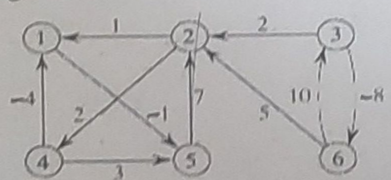
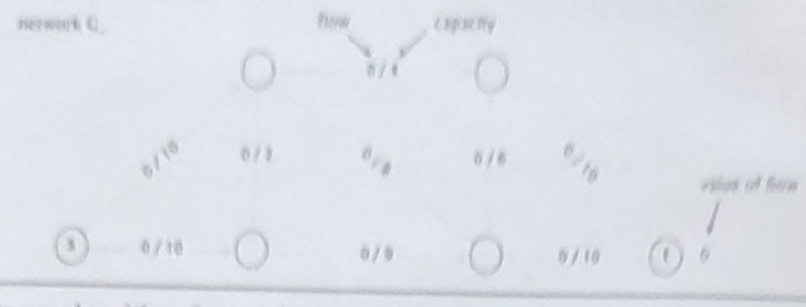
230
10
240

	RV College of Engineering® Department of Information Science and Engineering CIE - II : Test and Quiz Paper		
Subject : (Code)	ADVANCED ALGORITHMS (18IS6C2)	Semester :	6^{th} BE
Date : June 2022	Duration : (20+90) = 110 minutes Max. Marks: (10+50) = 60 Marks	Staff :RMS,KCG,TP,ASP	
Name :	USN :	Section :	IS/A/B/C

NOTE: *Answer all the questions from Part-A (10 M) and Part-B (50 M)*

Sl.no	PART – A (Quiz)	Marks	* BT	*CO
1.1	In Johnson's algorithm, the two important properties to be satisfied for the new set of edge weights w' are :____________ & ____________.	2	L2	CO2
1.2	From the given graph, how many vertices can be matched using maximum matching in bipartite graph algorithm? Mark them and list the edges. 	2	L2	CO2
1.3	The residual capacity of an edge(u,v) in a network with a flow f is given by__________. The residual capacity $C_f(u, v)$ = _______, if C(u,v)= 16, f(u,v)= 5.	2	L3	CO2
1.4	Given a flow network, let f be any flow and let (A,B) be any cut. Then, the net flow across (A,B) is __________ the value of f.	1	L1	CO1
1.5	The complexity of ________ sorting algorithm remains to be the same in its best, average and worst case?	1	L1	CO1
1.6	Why can't we use counting sort as a general purpose sorting algorithm? Give the performance analysis of Counting sort.	2	L2	CO3

Sl.no.	PART - B	Marks	* BT	*CO
2.	Use Johnson's algorithm to find the shortest paths between all pairs of vertices in the graph. Show the values of h and w' computed by the algorithm. 	10	L3	CO3

3.	For the following graph given below, find the augmented path and the maximum network flow using Ford-Fulkerson method. Draw the residual graph for each path and record all the calculations and steps 	10	L3	CO3
4.	Write an algorithm for sorting array elements using Bucket Sort. Use the above algorithm to sort the given array 22,45,12,8,10,6,72,81,33,18,50,14. Show the step wise sorting details.	10	L3	CO4
5a.	Demonstrate Counting sort algorithm with an example.	05	L2	CO2
5b.	Compare and Contrast Bucket sort algorithm and counting sort algorithm.	05	L2	CO2
6.	Consider the following problem of allocating students to classes. There are n students $s_1, s_2, s_3 \ldots, s_n$ and m classes $C_1, C_2, C_3, \ldots. C_m$. Each class C_j has its maximum enrollment capacity c_j. Each student s_i has a list L_i of 6 courses she/he would like to take during the next year. The university should enroll the student into 5 out of the 6 requested classes. The goal is to find enrollment of the students to classes that satisfies the following conditions: a) The number of students in each class C_j does not exceed the capacity c_j. b) Each student si is enrolled into exactly 5 classes from the list L_i. Show how to convert this problem into a Max-Flow problem on a flow network.	10	L4	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test + Quiz	Max Marks	02	11	27	20	02	11	37	10	-	-

Course Outcomes:

1. Analyze various algorithms for their time and space complexity
2. Demonstrate a familiarity with major algorithms and data structures
3. Apply appropriate design techniques for solving real world problems.
4. Design and implement solutions using appropriate mathematical techniques

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R V College of Engineering
Department of Computer Science and Engineering
CIE – II

Subject : (Code)	WEB TECHNOLOGY (18IS6D1)	Semester : 6^{TH} BE

S. No	PART-A	M	BT	Co
1.	Write script to be included in the document to produces a confirmation that the email with the given subject has been sent using alert function	2	2	3
2.	What is the role of the number 4 in the line: Alert(myArray[4])?	2	1	3
3.	Determine the number of elements, and their indices, in the array 'ages' created in the statement: var ages = new Array(3)	2	2	3

4.	Determine the output of the following code:
	
	var car = new Vehicle("Honda", "white", "2010", "UK");
	document.write(car);
	
	function Vehicle(model, color, year, country) {
	this.model = model;
	this.color = color;
	this.year = year;
	this.country = country; }
		2	2	3

5.	What is the output of the following code?
	var y = 1;
	if (function f() {}) {
	y += typeof f;
	}
	document.write(y);
		1	1	3

6.	Determine the output of below equality check? document.write(0.1 + 0.2 === 0.3);			
	PART-B	M	BT	Co
1.	Design a JavaScript program to calculate multiplication and division of two numbers (input form user). Sample Form: 1st Number : 12 2nd Number: 10 [Multiply] [Divide] The Result Is : 120	10	L4	3

2.	Design a JavaScript program to find two elements of the array such that their absolute difference is not greater than a given integer but is as close to the said integer. (Use functions)	10	L4	3
3a.	Define Constructors in JavaScript. Write a program to create a Car object with the properties namely Model, Make & Year and initialize the values for data properties. Display the details of Car object by initializing the display method.	5	L2	3
3b.	Demonstrate with a JavaScript code to validate the password field by considering the details given below. It is required to check for the following conditions explicitly 1. The password is entered using a prompt 2. The password should be at least 10 characters 3. Password should have at least 3 numbers and 2 special characters. 4. The password should be displayed as a * in the prompt. 5. The password should not contain spaces.	5	L4	3
4.	Write a javascript that inputs integers (one at a time) and passes them one at a time to function isEven, which uses the modulus operator to determine whether an integer is even. The function should take an integer argument and return true if the integer is even and false otherwise. Use sentinel-controlled looping and a prompt dialog.	10	L3	3
5	Write a XHTML file and JavaScript scripts to Find the Longest Word in a String. 1. Find the Longest Word With a FOR Loop 2. Find the Longest Word With the sort() Method	10	L3	2,3

COURSE OUTCOMES:

1. Understand the basic syntax and semantics of HTML/XHTML
2. Apply HTML/XHTML tags for designing static web pages and forms using Cascading Style Sheet.
3. Develop Client-Side Scripts using JavaScript and Server-Side Scripts using PHP and utilize the concepts of XML & Ajax to design dynamic web pages
4. Develop web based applications using PHP, XML and Ajax.

	L1	L2	L3	L4	L5	L6	CO1	CO2	CO3	CO4
Total Marks	3	11	20	25	-	-	-	5	55	-